# Atomic Capacity Does Not Determine Sublinear Causal Entropy Growth

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**Abstract**

We prove a realization theorem separating atomic capacity from sublinear finite-horizon simulation cost. For every unbounded increasing discretely concave sequence $f$ whose increments lie in $[0,1]$ and converge to zero, and every $a \in [0,1]$, there exists a one-action process with at most three output symbols per step such that

$$\mathrm{atomQ} = a, \qquad C_n \sim f(n).$$

The construction gives an exact finite-horizon entropy identity and a logarithmic additive error bound. Consequently, even the leading asymptotic of a diverging sublinear simulation cost does not determine whether an exact countably supported infinite-horizon seed exists. Moreover, the conclusion $C_n = o(n)$ for countably realizable processes admits no universal sublinear upper rate. Historical priority and Lean formalization are not asserted.

## 1 Definitions

All logarithms and entropies are in base two. Let $X = (X_1, X_2, \ldots)$ be a process on a fixed finite output alphabet, and write

$$P_n = \mathcal{L}(X_1, \ldots, X_n).$$

We regard $X$ as a controlled tree with a single available action.

Let $C_n$ be the infimum of the Shannon entropy of a discrete seed which, through deterministic decoding, realizes $P_n$ exactly. In this one-action setting,

$$C_n = H(P_n). \tag{1}$$

Indeed, entropy cannot increase under a deterministic map, while the transcript itself is an admissible seed.

For a finite probability law $P$, let $A_K(P)$ be the total mass of its $K$ largest atoms, using all atoms if fewer than $K$ exist. Define

$$\mathrm{atomQ}(X) = \sup_{K \geq 1} \inf_{n \geq 1} A_K(P_n). \tag{2}$$

The constructions below directly verify this quantity. They do not require an imported atomic-capacity identification theorem.

Deterministic transitions are permitted. Equivalently, zero-probability output branches may be deleted. No stationarity assumption is imposed.

## 2 The realization theorem

**Theorem 1** (Independent prescription of capacity and scale)**.** *Let $f : \mathbb{N}_0 \to [0, \infty)$ satisfy $f(0) = 0$ and*

$$q_n := f(n) - f(n-1), \qquad 1 \geq q_1 \geq q_2 \geq \cdots \longrightarrow 0, \qquad f(n) \longrightarrow \infty.$$

*For every $a \in [0, 1]$, there exists a one-action process on an alphabet of size at most three such that*

$$\boxed{\mathrm{atomQ}(X) = a, \qquad C_n \sim f(n).}$$

*More precisely, let $L$ be a nonnegative integer-valued random variable with*

$$\mathbb{P}(L \geq k) = q_k, \qquad k \geq 1.$$

*The process can be chosen so that, for every $n \geq 0$,*

$$\boxed{C_{n+1} = h_2(a) + a\big[f(n) + H(L \wedge n)\big] + (1-a)\lfloor f(n) \rfloor,} \tag{3}$$

*where $h_2$ denotes binary entropy. In particular,*

$$-1 \leq C_{n+1} - f(n) \leq 1 + a \log_2\big(e(f(n)+1)\big). \tag{4}$$

We first record the elementary entropy estimate used in the proof.

**Lemma 1** (Entropy controlled by the mean)**.** *If $Z$ is a nonnegative integer-valued random variable with finite mean $\mu$, then*

$$H(Z) \leq (\mu+1)\log(\mu+1) - \mu \log \mu \leq \log\big(e(\mu+1)\big),$$

*with $0 \log 0 = 0$.*

*Proof.* For $\mu = 0$, the variable is deterministic. For $\mu > 0$, compare the law of $Z$ with the geometric law

$$g_k = \frac{1}{\mu+1}\left(\frac{\mu}{\mu+1}\right)^k, \qquad k \geq 0.$$

The entropy–cross-entropy inequality gives

$$H(Z) \leq -\mathbb{E} \log g_Z = \log(\mu+1) + \mu \log\left(1 + \frac{1}{\mu}\right).$$

This inequality also holds for countable laws by the countable form of Gibbs' inequality, or by finite truncation. Finally,

$$\mu \log_2(1 + 1/\mu) \leq \log_2 e,$$

which proves the second bound. □

### 2.1 An atomic realization

Define $L$ by

$$\mathbb{P}(L = 0) = 1 - q_1, \qquad \mathbb{P}(L = k) = q_k - q_{k+1} \quad (k \geq 1).$$

These masses are nonnegative and sum to one because $q_k \to 0$. In particular, $L < \infty$ almost surely.

Conditional on $L$, choose a uniform binary word $(U_1, \ldots, U_L)$ and output

$$Y = (U_1, \ldots, U_L, \dagger, \dagger, \ldots).$$

The entire process is a deterministic function of the countable seed

$$S_{\mathrm{at}} = (L, U_1, \ldots, U_L).$$

**Lemma 2.** *The process $Y$ satisfies*

$$\mathrm{atomQ}(Y) = 1, \qquad H(Y_1, \dots, Y_n) = f(n) + H(L \wedge n).$$

*Consequently,*

$$f(n) \le H(Y_1, \dots, Y_n) \le f(n) + \log\bigl(e(f(n)+1)\bigr).$$

*Proof.* The set of trajectories

$$\bigcup_{\ell \ge 0} \{0,1\}^\ell \times \{\dagger^\infty\}$$

is countable.

For every $\delta > 0$, finitely many of these trajectories carry total mass at least $1 - \delta$. If their number is $K$, their prefixes occupy at most $K$ atoms at every horizon. Thus

$$A_K(\mathcal{L}(Y_1, \dots, Y_n)) \ge 1 - \delta \quad \text{for every } n.$$

Definition (2) gives $\mathrm{atomQ}(Y) = 1$.

The length-$n$ transcript determines $L \wedge n$. Conditional on $L \wedge n = k$, the transcript contains precisely $k$ independent fair bits and, when $k < n$, a deterministic suffix of stopping symbols. Hence

$$H(Y_1, \dots, Y_n) = H(L \wedge n) + \mathbb{E}(L \wedge n).$$

The tail-sum identity yields

$$\mathbb{E}(L \wedge n) = \sum_{k=1}^{n} \mathbb{P}(L \ge k) = \sum_{k=1}^{n} q_k = f(n).$$

Apply Lemma 1. □

## 2.2 A nonatomic realization

Put

$$k_n = \lfloor f(n) \rfloor, \qquad k_0 = 0.$$

Since $0 \le q_n \le 1$, each difference $k_n - k_{n-1}$ belongs to $\{0, 1\}$.

Define a binary process $Z$ as follows: at time $n$, output a fresh independent fair bit if $k_n - k_{n-1} = 1$, and otherwise output a deterministic zero.

**Lemma 3.** *The process $Z$ satisfies*

$$\mathrm{atomQ}(Z) = 0, \qquad H(Z_1, \dots, Z_n) = \lfloor f(n) \rfloor.$$

*Its infinite-path law is nonatomic.*

*Proof.* The length-$n$ law is uniform on $2^{k_n}$ transcripts. Therefore

$$H(Z_1, \dots, Z_n) = k_n, \qquad A_K(\mathcal{L}(Z_1, \dots, Z_n)) \le K 2^{-k_n}.$$

Since $k_n \to \infty$, definition (2) gives $\mathrm{atomQ}(Z) = 0$.

Every infinite trajectory specifies infinitely many independent fair bits, and therefore has probability zero. □

### 2.3 Mixing the two realizations

*Proof of Theorem 1.* At the first time, output a branch indicator $B$ with

$$\mathbb{P}(B=0)=a, \qquad \mathbb{P}(B=1)=1-a.$$

Conditional on $B=0$, continue with $Y$. Conditional on $B=1$, continue with $Z$. The total alphabet is $\{0,1,\dagger\}$.

Because the branch is explicitly observed, the chain rule, (1), and Lemmas 2–3 give

$$C_{n+1}=h_2(a)+a\big[f(n)+H(L\wedge n)\big]+(1-a)\lfloor f(n)\rfloor.$$

We check capacity directly. For any $\delta>0$, choose finitely many trajectories of $Y$ with total conditional mass at least $1-\delta$. Their mixture trajectories have total mass at least $a(1-\delta)$ and occupy a uniformly bounded number of prefix atoms. Hence $\mathrm{atomQ}(X)\geq a$.

Conversely, for every fixed $K$, any $K$ prefix atoms at horizon $n+1$ have total mass at most

$$a+K(1-a)2^{-k_n}.$$

Letting $n\to\infty$ gives $\mathrm{atomQ}(X)\leq a$. Thus $\mathrm{atomQ}(X)=a$.

Subtracting $f(n)$ from (3),

$$C_{n+1}-f(n)=h_2(a)+aH(L\wedge n)+(1-a)(\lfloor f(n)\rfloor-f(n)).$$

Since $0\leq h_2(a)\leq 1$ and $-1<\lfloor f(n)\rfloor-f(n)\leq 0$, Lemma 1 proves (4).

As $f(n)\to\infty$,

$$\frac{\log(f(n)+1)}{f(n)}\longrightarrow 0.$$

Therefore $C_{n+1}/f(n)\to 1$. Finally, $0\leq f(n+1)-f(n)\leq 1$ implies $f(n+1)/f(n)\to 1$, proving $C_n\sim f(n)$. □

## 3 Consequences

**Corollary 1** (The leading scale does not identify countability)**.** *For every $f$ satisfying Theorem 1, there exist two processes with*

$$C_n\sim f(n)$$

*such that one admits an exact countably supported seed for its entire infinite trajectory, while the other admits no such seed.*

*Proof.* Take $Y$ and $Z$.

The seed $(L,U_1,\dots,U_L)$ realizes $Y$ on a countable alphabet. Its entropy is infinite because

$$H(S_{\mathrm{at}})\geq \mathbb{E}L=\lim_{n\to\infty} f(n)=+\infty.$$

In fact, every exact seed for $Y$ has infinite entropy: its entropy is at least $C_n$ for every $n$.

The infinite-path law of $Z$ is nonatomic. A deterministic image of a countably supported random variable has countable support, so no countably supported seed can realize $Z$. □

The next result makes the absence of a universal convergence rate precise.

**Corollary 2** (No universal sublinear upper rate)**.** *For every function $b : \mathbb{N} \to (0, \infty)$ satisfying $b(n) = o(n)$, there exists a countably realizable one-action process with at most three outputs per step such that*

$$\boxed{\text{atomQ} = 1, \qquad \frac{C_n}{n} \longrightarrow 0, \qquad \frac{C_n}{b(n)} \longrightarrow +\infty.}$$

*Proof.* Set

$$r(n) = \sqrt{n \max\{b(n), 1\}}, \qquad q_k = \min\left\{1, \sup_{j \geq k} \frac{r(j)}{j}\right\}, \qquad f(n) = \sum_{k=1}^{n} q_k.$$

Since $b(n) = o(n)$, we have $r(n)/n \to 0$. Consequently $q_k$ is nonincreasing, belongs to $[0, 1]$, and converges to zero.

For all sufficiently large $n$, $r(n)/n \leq 1$. For every $k \leq n$, the supremum defining $q_k$ includes $j = n$, so

$$q_k \geq \frac{r(n)}{n}.$$

Thus $f(n) \geq r(n)$ for all sufficiently large $n$. In particular, $f(n) \to \infty$.

Apply the atomic construction of Lemma 2. It gives

$$C_n \geq f(n) \geq r(n).$$

If $b(n) \geq 1$, then

$$\frac{r(n)}{b(n)} = \sqrt{\frac{n}{b(n)}} \longrightarrow \infty;$$

if $b(n) < 1$, then $r(n)/b(n) \geq \sqrt{n}$. Hence $C_n/b(n) \to \infty$.

On the other hand, Cesàro convergence gives $f(n)/n \to 0$ because $q_k \to 0$. Using $f(n) \leq n$ and Lemma 2,

$$0 \leq \frac{C_n}{n} \leq \frac{f(n)}{n} + \frac{\log(e(n+1))}{n} \longrightarrow 0.$$

□

## 4 Relation to the causal capacity–rate bound

For bounded output alphabets, the causal capacity–rate framework gives the constraint

$$R \leq 1 - \text{atomQ},$$

where $R$ is the normalized upper linear growth rate of finite-horizon seed entropy.

Theorem 1 describes a different aspect of the zero-rate region. Throughout the stated class of concave sublinear scales, every atomic capacity is compatible with the same leading entropy asymptotic:

$$\big(\text{atomQ},\ C_n/f(n)\big) \longrightarrow (a, 1) \qquad \text{for every } a \in [0, 1].$$

Thus the sublinear growth scale alone cannot recover the atomic capacity.

Corollary 2 also shows that the implication

$$\text{exact countable realization} \implies C_n = o(n)$$

cannot be strengthened to a universal prescribed sublinear upper rate, even in the one-action subclass.

These statements concern the scalar capacity atomQ. They do not assert that the examples have the same limiting information envelope $F_\infty$, nor do they determine how much additional temporal information that full profile carries.

## 5 Scope and verification status

The proofs are self-contained apart from standard entropy identities and Gibbs' inequality. The construction permits deterministic transitions and is generally nonstationary.

The theorem realizes a broad class of prescribed growth scales; it is not a classification of every possible entropy sequence. It does not assert analogous freedom for stationary kernels or for fixed strictly positive transition probabilities.

The mechanisms used in the proof—random finite words, scheduled independent bits, and distinguishable mixtures—are elementary. Historical novelty of their precise realization consequence requires a separate literature assessment. No Lean-kernel verification is claimed.

## References


[1] J. Mikulík, *One Seed for Every Adaptive Policy*. Mathematical notes and Lean development. `https://github.com/jendamikulik/finite-Shannon-theorem`.

[2] J. Mikulík, *Atomic Capacity and the Entropy Rate of Causal Simulation: A Sharp Feasible-Region Theorem for Bounded Output Alphabets*. Research note, 20 October 2026.